TASK SPECIFICATION: P5-03 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Parking Reservation Engine & Digital Pass Generation

Git Branch: `feature/P5-03-parking-booking` · Target Pull Request: feature/P5-03-parking-booking

> **Core Objective & Technical Scope:**
> Collision-free spot reservation backed by PostgreSQL GiST exclusion constraints, server-side tariff calculation, and encrypted QR pass.

Priority Level	Critical	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-03-parking-booking`
Upstream Dependencies	P5-02, P4-02		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V7__booking.sql migration applying btree_gist exclusion constraint on (spot_id, booking_interval).
2. Implement PricingService computing exact tariff based on vehicle type and duration.
3. Wire reservation creation with atomic wallet debit in single transaction.
4. Generate encrypted pass payload with QR verification token.
5. Execute multi-threaded concurrency tests to verify collision prevention.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V7__booking.sql`
- `parking/application/Booking*`
- `parking/api/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Double-booking mathematically impossible under concurrent execution.
- Payment failure rolls back reservation atomically.

VERIFICATION & TEST COVERAGE STRATEGY

- PricingServiceTest
- ConcurrencyBookingStressTest
- BookingControllerTest